Mini-Project

PCA vs. Truncated SVD for Retail Recommendations

Goal: better Top-10 product recommendations and useful customer groups

1. Data preparation

- 1,067,371 raw transaction rows across two Excel sheets
- Removed cancellations, invalid prices/quantities, non-product codes and missing customer IDs
- 802,920 cleaned rows; last two months held out for testing
- Training matrix: 5,475 customers × 3,689 products

97.95% sparse • CSR 4.76 MiB vs. dense 154.09 MiB

2. Why Truncated SVD?

- Customer–product data contains mostly zeros
- Truncated SVD works directly on sparse matrices without mean centering
- 50 components represent customer and product purchase patterns
- PCA tested on the same data for comparison

At k = 50: 34.88% explained variance; 78.94% relative reconstruction error

3. Recommendation results

Hit Rate@10: at least 1 matching held-out item in 10 suggestions

Method	Hit Rate@10
Best-seller baseline	17.41%
SVD k=50 (Log)	37.32%
SVD k=100 (Log)	38.06%
SVD k=50 (Binary)	40.15%

Evaluated on 1,913 eligible customers; already purchased products excluded.

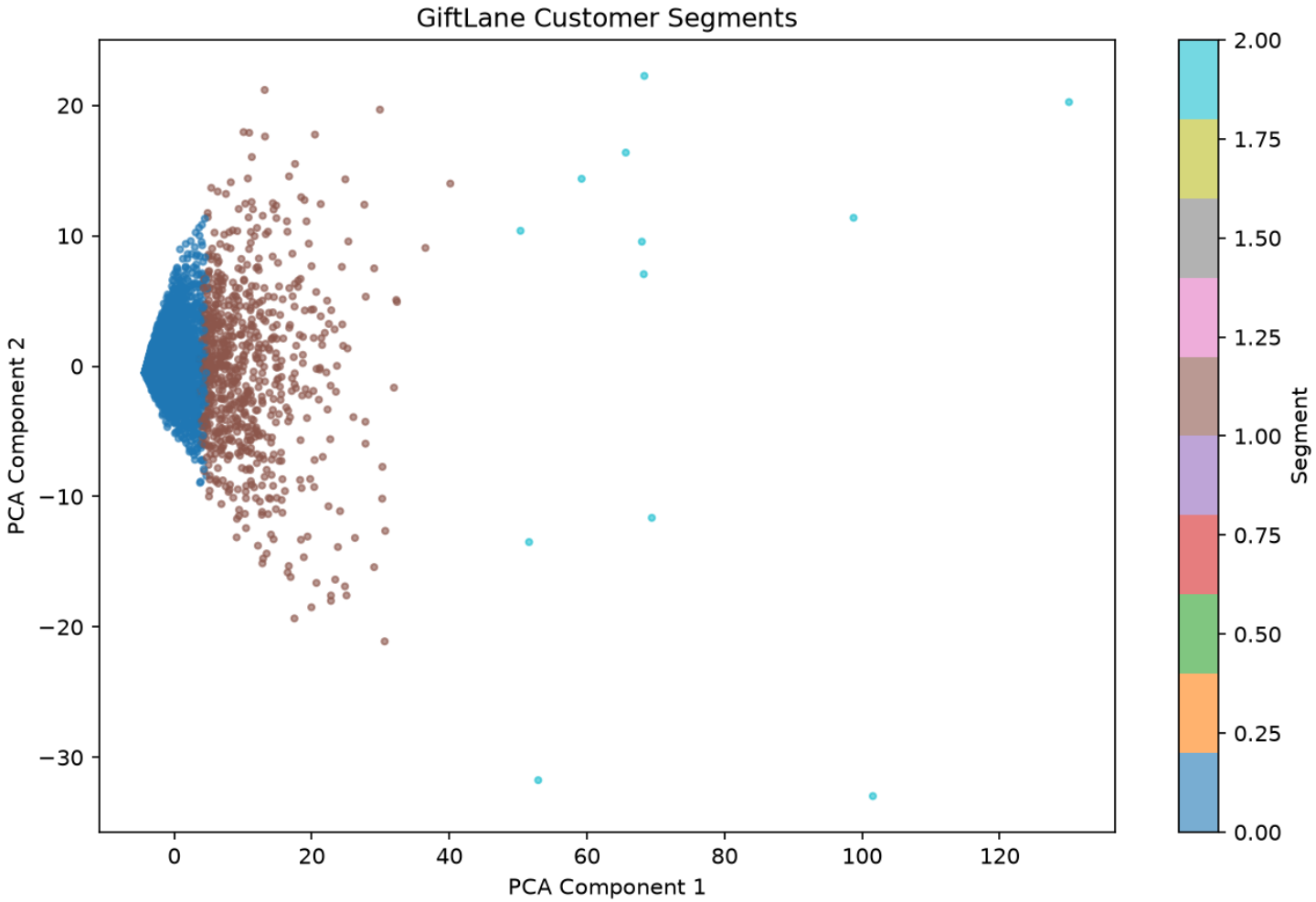
4. PCA versus SVD (k = 50, Log)

Measure	PCA	SVD
Fit time	0.1021 s	0.1057 s
Stored factors	3.524 MiB	3.496 MiB
Peak process memory	207.92 MiB	234.64 MiB
Hit Rate@10	37.27%	37.32%

Performance is almost identical; SVD was kept for sparse-input compatibility.

Peak process memory includes runtime overhead.

5. Customer groups



3 clusters (K-Means on SVD embeddings)

0 Home & Giftware

4,734 customers · 86.47%

1 Retrospot & Everyday

729 customers · 13.32%

2 Lunchware & Tableware

12 customers · 0.22%

Group names are preliminary.

6. Conclusion and next steps

- SVD recommendations outperform the best-seller baseline
- Binary purchase values performed best among tested k=50 options
- PCA and SVD show nearly identical recommendation accuracy
- Next: test more k values and address cold-start customers/products

Limits: eligible historical purchases only; measured runtime excludes training.